

Rahul Puthukkot

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience designing and delivering scalable service platforms, cloud-native microservices, and data-driven systems in high-performance environments. Strong background in Node.js, TypeScript, Python, and API-driven architectures, with end-to-end ownership of backend platforms from design through production. Experienced in mentoring engineers, modernising legacy systems, and collaborating cross-functionally to deliver reliable, customer-facing products. Proactive in experimenting with emerging technologies and pragmatically integrating them into production to improve scalability, reliability, and engineering efficiency.

CORE SKILLS

Backend & API's: Node.js, TypeScript, Python, REST APIs, GraphQL, Microservices Architecture

Cloud & Infrastructure: Azure, Docker, Kubernetes, Terraform, CI/CD pipelines, Infrastructure as Code

Databases & Data : MongoDB, PostgreSQL, SQL, Neo4J, Data Modelling, Observability

Frontend: React.js, Next.js

AI/ML: RAG, Agent based workflows, LLM, Data Analysis

Engineering Leadership: Technical Ownership, Mentoring & Code Reviews, System Design, Cross-team collaboration, Agile/Scrum, Stakeholder Management

PROFESSIONAL EXPERIENCE

VC TECHNOSYS LTD

Coventry, UK

Senior Software Developer

2021-Present

McLaren Automotive

- Architected and deployed cloud-based ECU validation systems using Python (AsyncIO, FastAPI), Docker, and Azure, enabling scalable, real-time data ingestion and analysis for high throughput systems/
- Designed and implemented microservice based architectures to support automated testing, deployment workflows and rapid prototyping across multiple internal platforms
- Established production grade observability by integrating tracing and metrics collection, enabling proactive alerting, faster root cause analysis and improved system reliability
- Built monitoring dashboards to visualise health, performance, event logging and operation metrics, significantly improving transparency and incident response.
- Delivered internal Platform-as-a-Service tooling to standardise service scaffolding, CI/CD pipelines which reduced time to production.
- Designed and implemented AI-assisted information extraction using graph databases and Retrieval Augmented Generation(RAG), improving requirements analysis by 25%.
- Collaborate closely with product, platform and infrastructure stakeholders to gather requirements, design, develop and deliver scalable production ready services.

Aston Martin Lagonda

- Designed and built backend services to collect, process and synchronise high volume log and event data with cloud platforms, improving visibility and reducing issue resolution by 40%.
- Developed and maintained RESTful API's to integrate data collection tools, cloud storage, and validation platforms, enabling automated data pipelines and real time analysis.
- Led the development of reusable backed validation framework, standardising process across multiple teams, and improving consistency and maintainability.

- Engineered asynchronous and multi threading backend services to enable parallel processing, reducing end to end processing by 20%.
- Built simulation and test tooling to improve the reproducibility of complex scenarios and reduce reliance of physical infrastructure.
- Acted as a technical lead within the engagement, guiding design decisions, reviewing code, and supporting engineers through complex problem solving.

VC Technosys

- Developed and maintained backend driven automation frameworks to orchestrate large scale test and validation workflows, improving coverage and accelerating fault detection.
- Designed and delivered internal tooling to automate integration workflows, improving issue detection by 30% and increasing delivery confidence.
- Lead the technical interview process for backend, full-stack, and platform engineering roles, defining role-specific evaluation criteria, conducting system design and coding interviews, and providing structured, decisive recommendations that directly influence hiring decisions and team composition.
- Mentored engineers in asynchronous programming, test-driven development, and cloud-native deployment practices, contributing to improved code quality, architectural consistency, and overall technical maturity.
- Contributed to architectural discussions, tooling standards, and best practices adopted across multiple client engagements.

FREELANCE SOFTWARE DEVELOPER

London, UK

Software Engineer

2019-2021

- Delivered backend-heavy and full-stack solutions across healthcare, automation, and customer support platforms using Python, Django, FastAPI, and PostgreSQL.
- Built NLP-based chatbot integrated with CRM systems, automating ~70% of customer queries and reducing response times by 60%.
- Developed data monitoring, analytics, and visualisation tools to improve operational efficiency and decision-making.
- Designed scalable e-commerce and dashboard solutions with a focus on performance and maintainability..

QBURST TECHNOLOGIES

Trivandrum, India

Full Stack Developer

2016-2019

Fast Retailing

- Built and maintained backend services and APIs supporting high-traffic e-commerce and analytics platforms using Node.js, MongoDB, and PostgreSQL.
- Designed RESTful APIs and a GraphQL-based CMS to support dynamic content delivery and frontend-backend decoupling.
- Contributed to early microservices adoption and containerised backend services with Docker for cloud readiness.
- Implemented automated testing across backend and integration layers, reducing production defects by 30%.
- Collaborated with globally distributed teams in agile delivery environments.

EDUCATION

UNIVERSITY OF HERTFORDSHIRE

2021, United Kingdom

MSc Artificial Intelligence and Robotics

MAR ATHANASIOS COLLEGE OF ENGINEERING

2016, Kerala, India

Bachelors in Computer science and Engineering

Certifications

Neo4j Certified Professional

Microsoft Certified: Azure Developer Associate (In progress)